Node.js + React practice task project, written in a way that can be used for **real-world development, portfolio, or interview explanation**.

🔐 Authentication & Authorization System

1. User Authentication

Implement a complete authentication system with the following features:

🔑 Authentication Flows

User Registration

- Email & password-based signup
- Password hashing using **bcrypt**

User Login

- Issue **JWT Access Token** (short-lived)
- Issue **JWT Refresh Token** (long-lived)

Forgot Password

- Email-based password reset
- Secure reset token with expiry

🔄 Token Management

Access Token

- Used for API authorization
- Short expiry (e.g., 15 minutes)

Refresh Token

- Stored securely in DB

Used to generate new access token

Auto Logout

Logout user when refresh token expires or becomes invalid

🛡️ Security Enhancements

Role-Based Access Control (RBAC)

Roles: `Admin`, `Super Admin`, `User`

Protected Routes

Middleware to validate JWT

Middleware to validate user roles

Rate Limiting

Prevent excessive login attempts

Brute-Force Protection

Lock account temporarily after multiple failed login attempts

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🔐 OAuth Integration

Login with Google OAuth

OAuth2 flow using Google Identity

Auto-create user account if not exists

Assign default role (`User`)

🪪 KYC (Know Your Customer) System

2. Mandatory Multi-Step KYC (Post Login)

📋 KYC Flow

KYC is **mandatory after login**

Users **cannot access any page** until KYC is completed

Global route guard to check `kycStatus`

📄 Multi-Step KYC Form

Steps include:

Personal Details (Name, DOB, Address)

Identity Proof Upload (Aadhar / Passport)

Address Proof Upload

Review & Submit

📌 KYC Status Management

`Pending`

`Approved`

`Rejected`

Only users with **Approved KYC** can access the application.

🛒 Product Management System

3. Product Module

📦 Product Features

Create, update, delete products

Product image upload

Price, description, category, stock

🔐 Access Control

Only **Admin** and **Super Admin** can:----

Add products

Edit products

Soft delete products

🗑️ Soft Delete Implementation

Use `isDeleted: true`

Deleted products are hidden but retained in DB

Admin can restore deleted products

📦 Order Management System

4. Order Functionality

📜 Order Creation

Only **Users** can place orders

Orders linked to user and products

Order status lifecycle:

Pending

Confirmed

Shipped

Delivered

Cancelled

🔐 Order Access Rules

Users can view only **their own orders**

Admin can view and manage **all orders**

🗑️ Soft Delete Orders

Orders are soft deleted for auditing

Maintains order history

📂 File Handling & Cloud Storage

5. File Upload System

📤 Upload Features

File upload using **Multer**

Validate:

- File size
- File type (JPG, PNG, PDF)

Upload destination: **AWS S3 Bucket**

☁️ AWS S3 Integration

Secure S3 bucket configuration

Generate **signed URLs** for private file access

Store only file references in database

⚡ Performance Optimization (Frontend)

6. React Performance Improvements

🚀 Optimization Techniques

Code Splitting

`React.lazy()` and `Suspense`

Memoization

`useMemo` for heavy computations

`useCallback` for stable function references

Virtualized Lists

Efficient rendering for large data sets (orders, products)

🔐 Frontend Authentication & Route Protection

7. Secure Routing

🛡️ Protected Routes

Route guards based on:

Authentication

User role

KYC status

🔄 Token Handling

Automatic token refresh using refresh token

Silent re-authentication

Logout on token expiry or invalid refresh token

💬 Real-Time User-to-User Chat

8. Chat System (Socket.IO)

💬 Chat Features

One-to-one real-time messaging

Online/offline user status

Message delivery acknowledgments

🔐 Chat Security

Socket authentication using JWT

Prevent unauthorized socket connections

🗂️ Database & Architecture

9. Database Design

MongoDB with Mongoose

Soft delete across all collection

Indexed fields for performance

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10. Project Architecture

Modular folder structure

Centralized error handling

API versioning (`/api/v1`)

Logging and monitoring

✅ Tech Stack Summary

🚀

Layer	Technology
Backend	Node.js, Express
Database	MongoDB
Auth	JWT, OAuth
Frontend	React
State	Redux
Realtime	Socket.IO
Storage	AWS S3
Security	Rate Limit, RBAC